Vidyavardhini's College of Engineering &Technology
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Experiment No.3
Aim: Design and Configuration of a LAN Topology Using Hubs and Switches in Cisco Packet Tracer
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Experiment No 3

AIM: Design and Configuration of a LAN Topology Using Hubs and Switches in Cisco Packet Tracer

THEORY:

A Local Area Network (LAN) is a network that connects computers and other devices within a limited geographical area such as a classroom, laboratory, office, or building. LANs enable users to share resources like files, printers, applications, and Internet access while providing fast and reliable communication.

Cisco Packet Tracer is a network simulation software developed by Cisco that allows users to design, configure, and test network topologies in a virtual environment without using physical networking devices. In this experiment, a LAN topology is created using hubs and switches to understand their working, differences, and communication within a network.

Local Area Network (LAN)

- A LAN connects multiple computers within a small geographical area.
- It provides high-speed communication between connected devices.
- LANs are commonly used in schools, colleges, offices, and organizations.
- Devices in a LAN communicate using Ethernet technology.

Cisco Packet Tracer

- Cisco Packet Tracer is a network simulation and learning tool.
- It allows users to design network topologies using routers, switches, hubs, PCs, servers, and other networking devices.
- It provides both real-time and simulation modes to observe packet transmission.
- It is widely used for learning networking concepts and testing network configurations.

Hub

A Hub is a basic networking device that connects multiple computers in a LAN.

Features of Hub

- Operates at the Physical Layer (Layer 1) of the OSI model.
- Does not store MAC addresses.
- Broadcasts incoming data to all connected devices.
- Cannot identify the intended destination device.

- Shared bandwidth among all connected devices.
- Supports only half-duplex communication.
- Higher chances of network collisions.

Working of Hub

When a computer sends data to a hub, the hub simply forwards the data to every connected device. Every device receives the data, but only the destination device accepts it while the others discard it.

Switch

A Switch is an intelligent networking device used to connect multiple devices within a LAN.

Features of Switch

- Operates at the Data Link Layer (Layer 2) of the OSI model.
- Maintains a MAC Address Table.
- Sends data only to the intended destination device.
- Reduces unnecessary network traffic.
- Supports full-duplex communication.
- Minimizes network collisions.
- Provides better performance than a hub.

Working of Switch

A switch learns the MAC addresses of connected devices. When data arrives, it checks its MAC address table and forwards the frame only to the correct destination port instead of broadcasting it to all devices.

PROCEDURE AND OUTPUT:

Topology 1: Star Topology Using a Switch

Step 1: Device Placement

Six PCs (PC1–PC6) were placed around a single Cisco 2960-24TT switch (Switch0) to form a star topology, with the switch acting as the central connecting device.

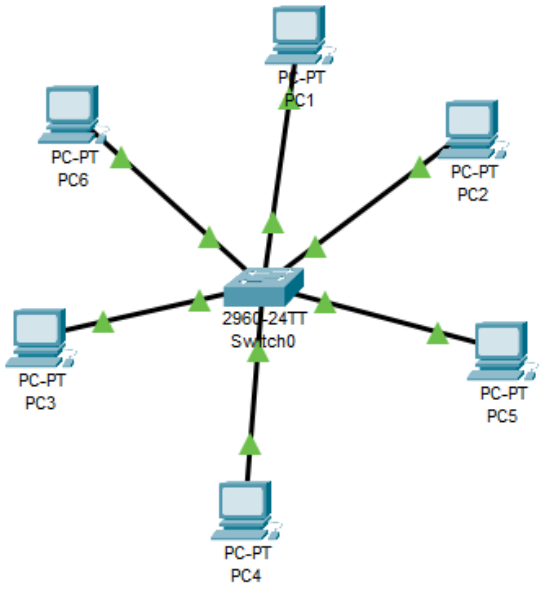


Figure 1: Star topology built with a switch (Switch0) connecting PC1–PC6.

Step 2: Cabling

Each PC's FastEthernet port was connected to a port on Switch0 using a Copper Straight-Through cable.

Step 3: IP Addressing Plan

PC	IP Address	Subnet Mask
PC1	192.168.1.1	255.255.255.0
PC2	192.168.1.2	255.255.255.0
PC3	192.168.1.3	255.255.255.0
PC4	192.168.1.4	255.255.255.0
PC5	192.168.1.5	255.255.255.0
PC6	192.168.1.6	255.255.255.0

Since all six PCs sit in a single switched broadcast domain, no router or default gateway is required for them to communicate with one another.

Step 4: Testing

Connectivity was verified by pinging between PCs across the switch. All tests returned 0% packet loss.

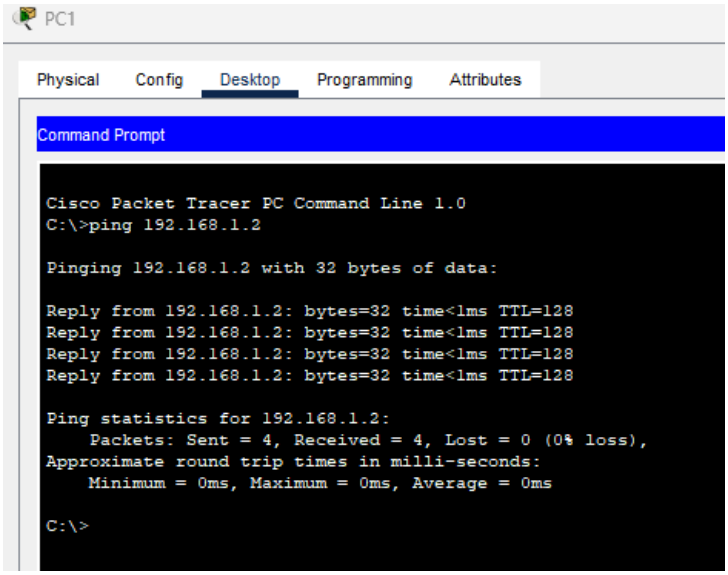


Figure 2: PC1 pinging PC2 (192.168.1.2), successful, 0% loss.

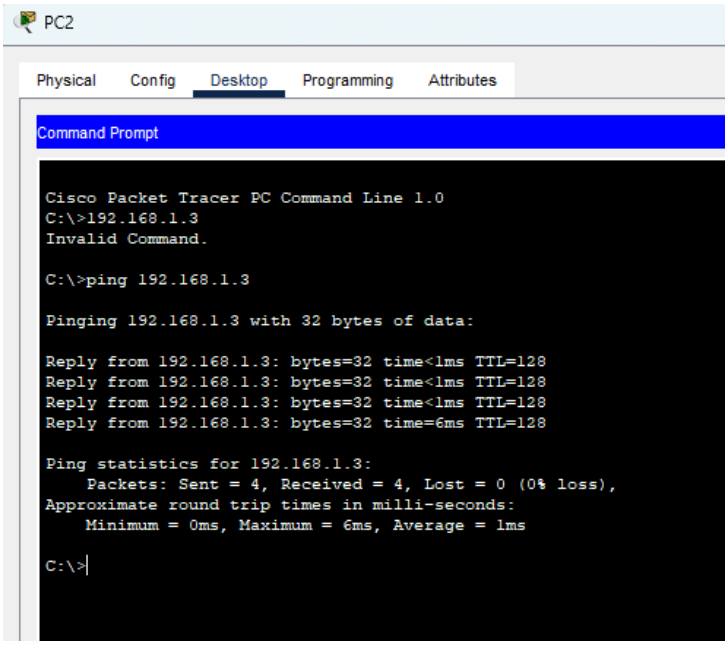


Figure 3: PC2 pinging PC3 (192.168.1.3), successful, 0% loss.

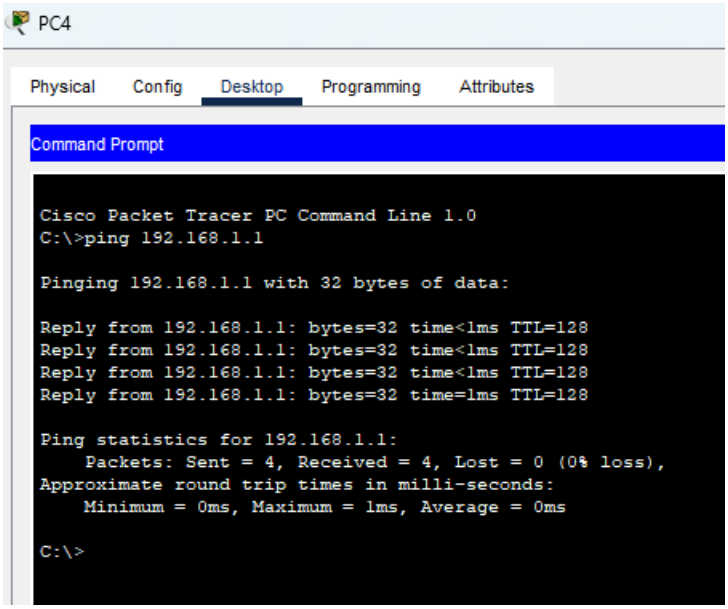


Figure 4: PC4 pinging PC1 (192.168.1.1), successful, 0% loss.

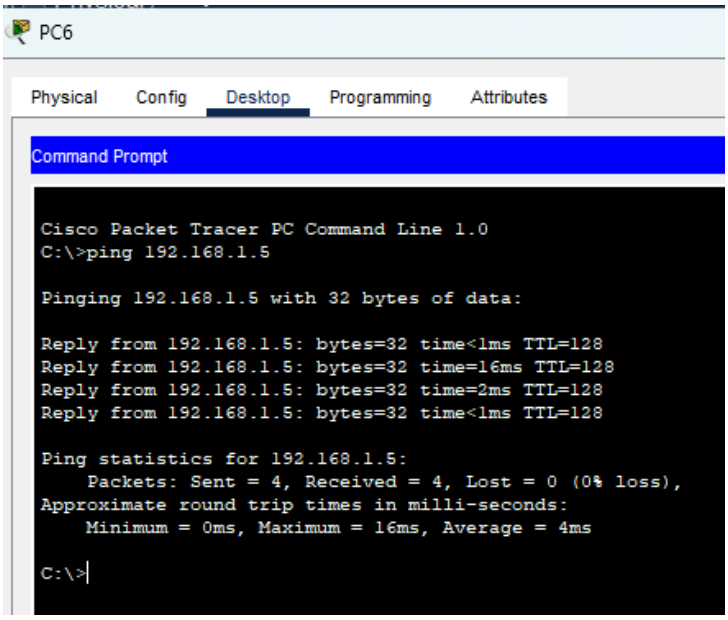


Figure 5: PC6 pinging PC5 (192.168.1.5), successful, 0% loss.

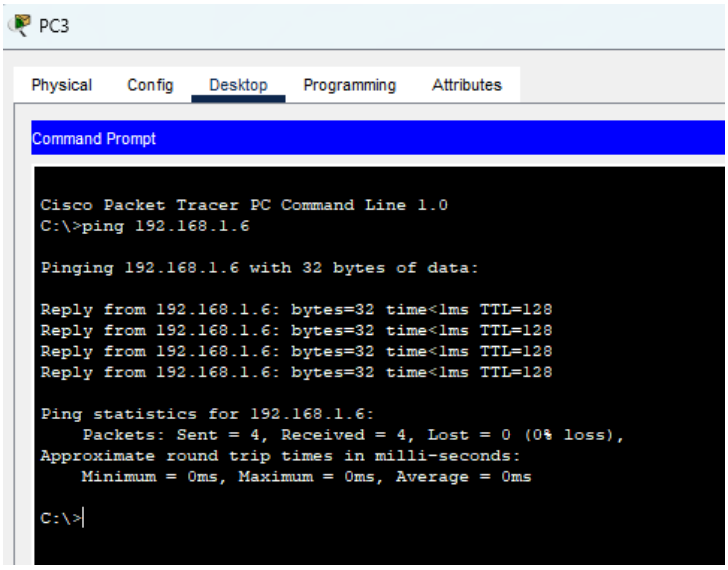


Figure 6: PC3 pinging PC6 (192.168.1.6), successful, 0% loss.

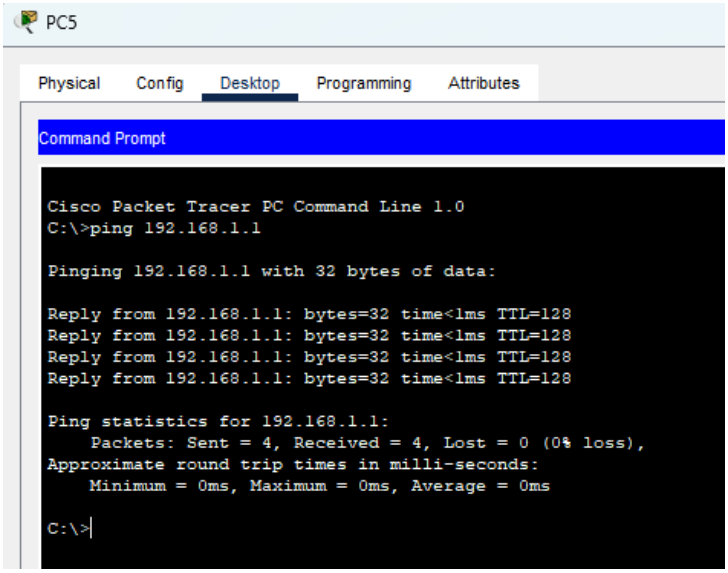


Figure 7: PC5 pinging PC1 (192.168.1.1), successful, 0% loss.

Observation:

All ping tests succeeded with 0% packet loss and low, fairly consistent round-trip times (0–6 ms). In Simulation mode, the switch delivered each frame directly to the destination port only, confirming that Switch0 was learning MAC addresses and forwarding intelligently rather than broadcasting.

Topology 2: Star Topology Using a Hub

Step 1: Device Placement

The same six PCs (PC1–PC6) were arranged in a star layout, but this time connected to a hub (Hub0) instead of a switch, so that hub and switch behaviour could be compared under an identical physical layout.

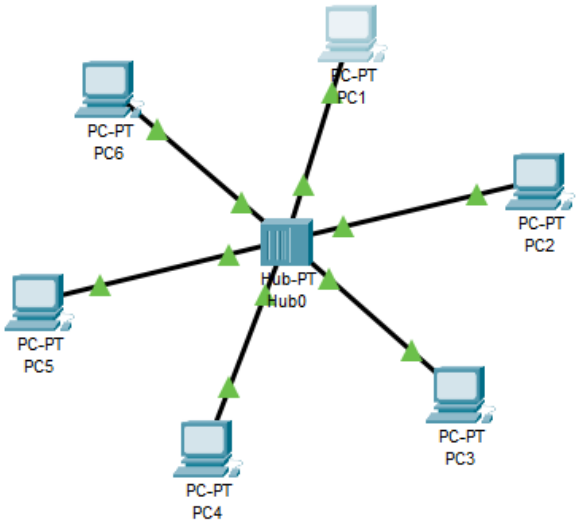


Figure 8: Star topology built with a hub (Hub0) connecting PC1–PC6.

Step 2: Cabling

Each PC's FastEthernet port was connected to a port on Hub0 using a Copper Straight-Through cable.

Step 3: IP Addressing Plan

The same addressing plan as Topology 1 (192.168.1.1 – 192.168.1.6)was reused for the hub-based star topology, so that the two topologies could be compared fairly.

Step 4: Testing

The same set of ping tests was repeated with the hub in place of the switch. All tests again returned 0% packet loss.

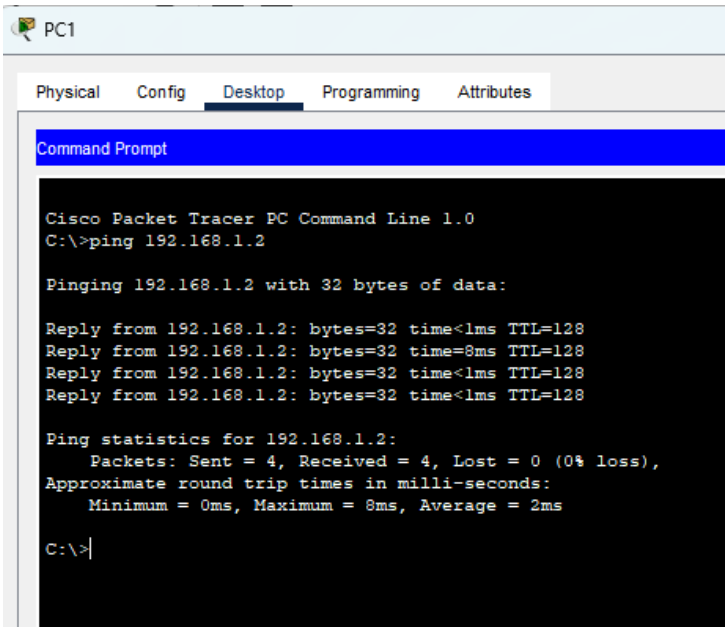


Figure 9: PC1 pinging PC2 (192.168.1.2) via the hub, successful, 0% loss.

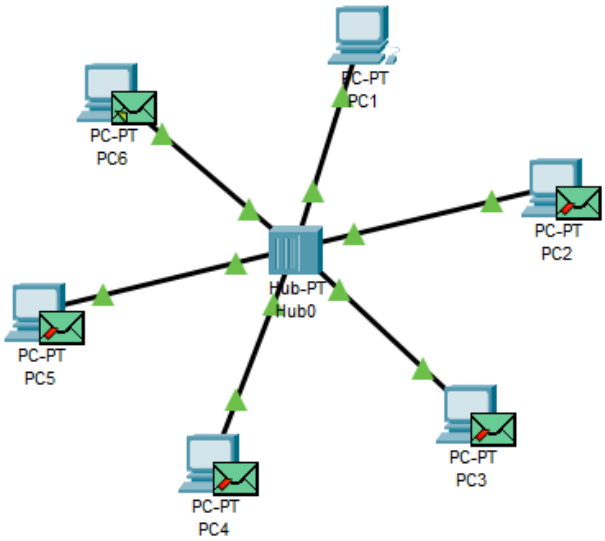


Figure 10: Hub0 flooding the ping request out to every connected PC (envelope icons at all six stations) instead of only the destination, visible in Simulation mode.

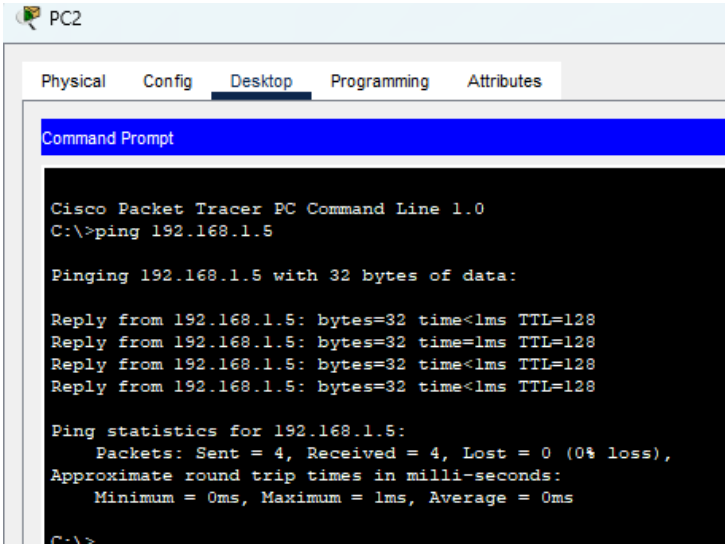


Figure 11: PC2 pinging PC5 (192.168.1.5) via the hub, successful, 0% loss.

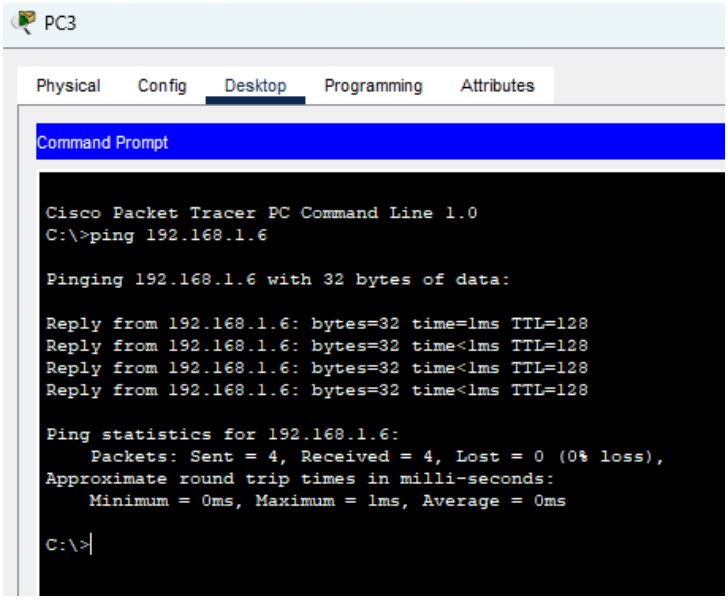


Figure 12: PC3 pinging PC6 (192.168.1.6) via the hub, successful, 0% loss.

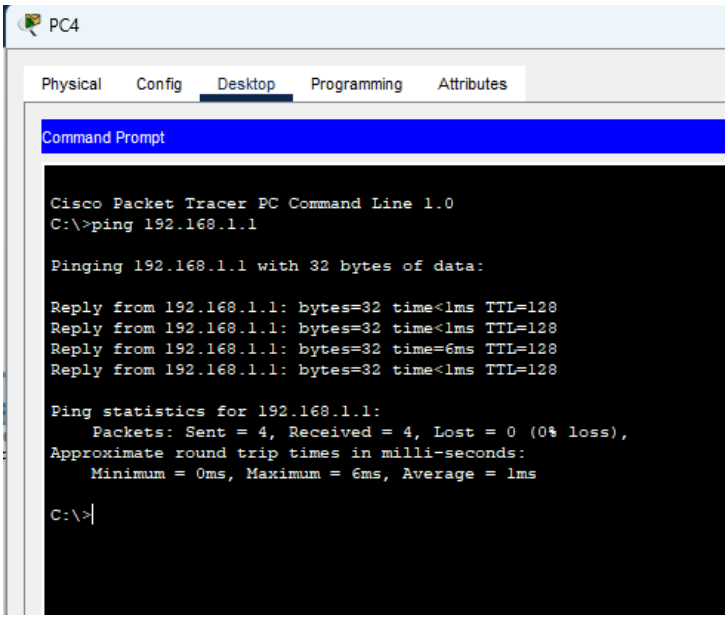


Figure 13: PC4 pinging PC1 (192.168.1.1) via the hub, successful, 0% loss.

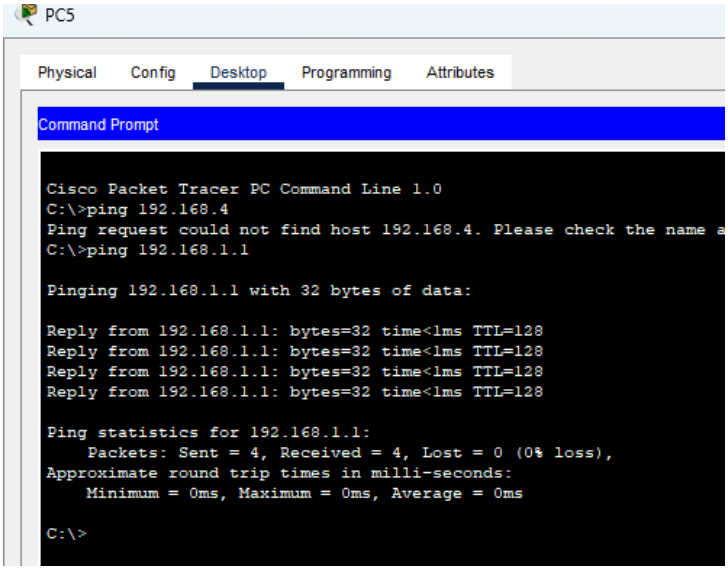


Figure 14: PC5 pinging PC1 (192.168.1.1) via the hub, successful, 0% loss (an initial incomplete address, 192.168.4, was corrected first).

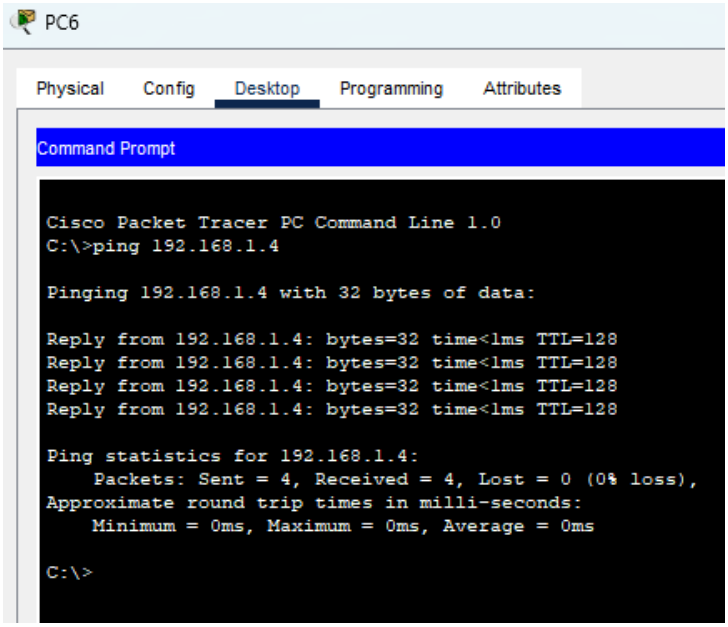


Figure 15: PC6 pinging PC4 (192.168.1.4) via the hub, successful, 0% loss.

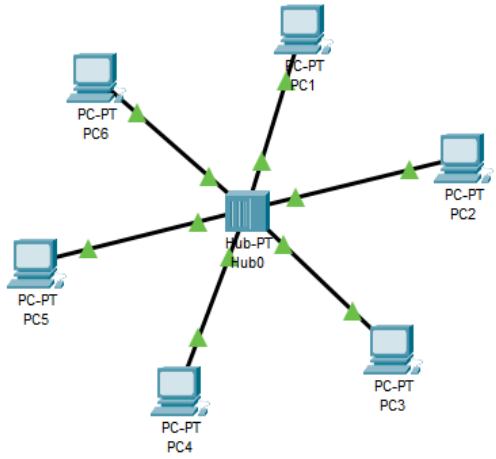


Figure 16: Cisco Packet Tracer workspace showing the completed star_topology_hub project file.

Observation:

Every ping again succeeded with 0% packet loss, but round-trip times were slightly higher and more variable (up to 8–16 ms maximum) than with the switch. Figure 10 shows Hub0 broadcasting the ping frame out of every port at once, all six PCs display an envelope icon even though only one of them is the actual destination, confirming the flood-to-all, shared-bandwidth behaviour described under "Working of Hub."

Topology 3: Bus Topology Using a Hub

Step 1: Device Placement

Six PCs (PC0–PC5) were laid out in a horizontal line to represent a bus-style layout, with every PC's cable run back to a shared hub (Hub0), using the hub as the common shared medium that a physical bus segment would otherwise provide.

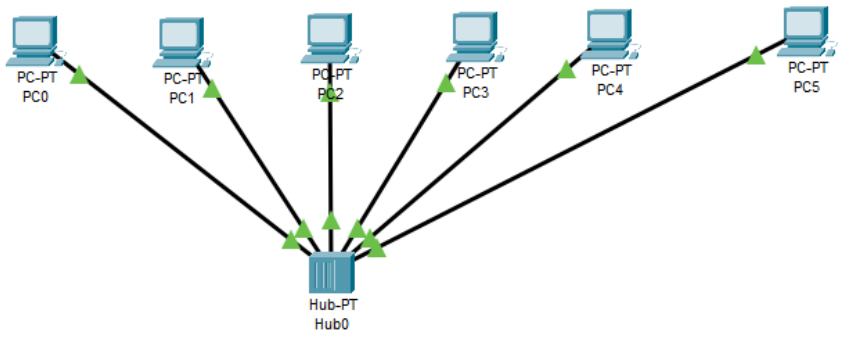


Figure 17: Bus-style layout of PC0–PC5 sharing a common hub (Hub0).

Step 2: Cabling

Each PC was connected to Hub0 with a Copper Straight-Through cable.

Step 3: IP Addressing Plan

PC	IP Address	Subnet Mask
PC0	192.168.2.1	255.255.255.0
PC1	192.168.2.2	255.255.255.0
PC2	192.168.2.3	255.255.255.0
PC3	192.168.2.4	255.255.255.0
PC4	192.168.2.5	255.255.255.0
PC5	192.168.2.6	255.255.255.0

Step 4: Testing

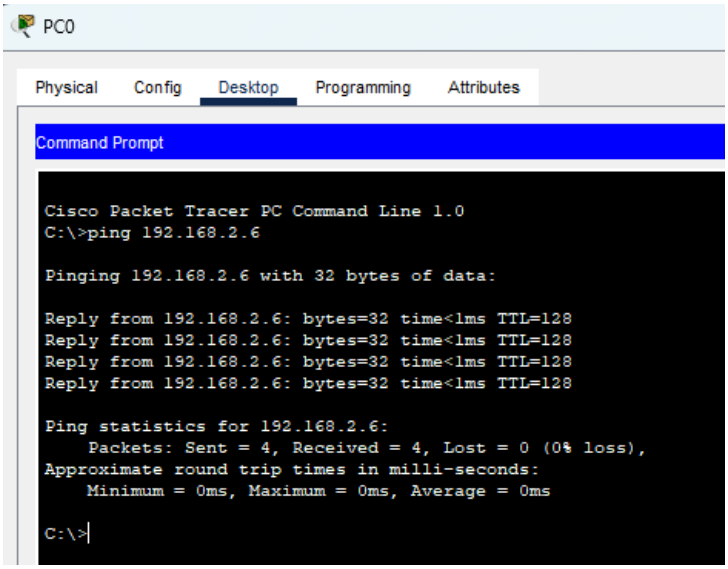


Figure 18: PC0 pinging 192.168.2.6 across the shared hub, successful, 0% loss.

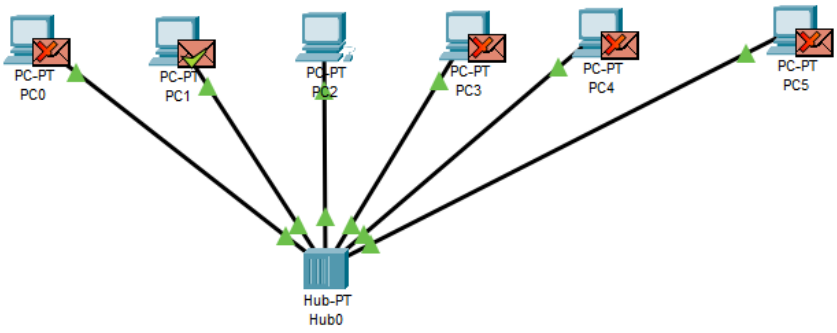


Figure 19: Simulation-mode view of the same ping, the frame is flooded to every station; a check mark shows successful delivery at the destination while the others show it being discarded, since they are not the intended recipient.

Observation:

The ping test succeeded with 0% packet loss. As with the star-hub topology, Figure 19 confirms that the hub floods every incoming frame to all connected stations regardless of the physical layout used to wire them, only the destination PC keeps the frame, while every other PC discards it.

Topology 4: Ring Topology Using Switches

Step 1: Device Placement

Six 2960-24TT switches (Switch0–Switch5) were arranged in a hexagonal ring, with each switch connected to its two neighbours to form a closed loop. One PC (PC0–PC5) was attached to each switch.

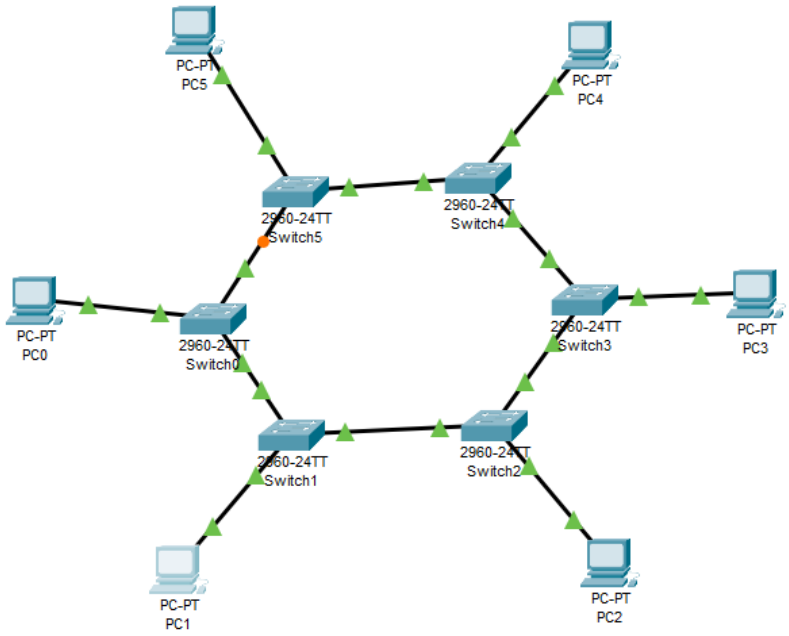


Figure 20: Ring topology, six switches (Switch0–Switch5) interconnected in a loop, each carrying one PC.

Step 2: Cabling

Copper Straight-Through cables were used between each PC and its switch, and between adjacent switches around the ring.

Step 3: IP Addressing Plan

All six PCs were addressed from the 192.168.3.0/24 network (192.168.3.1 – 192.168.3.6, subnet mask 255.255.255.0), one address per PC.

Step 4: Testing

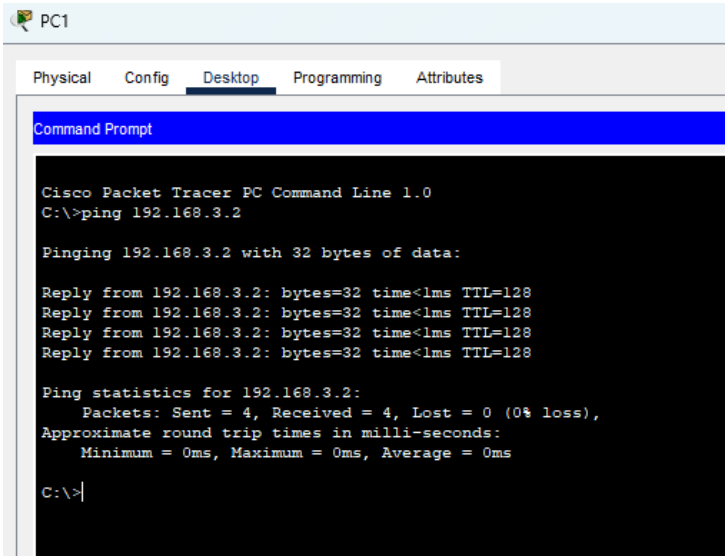


Figure 21: PC1 pinging 192.168.3.2 around the ring of switches — successful, 0% loss.

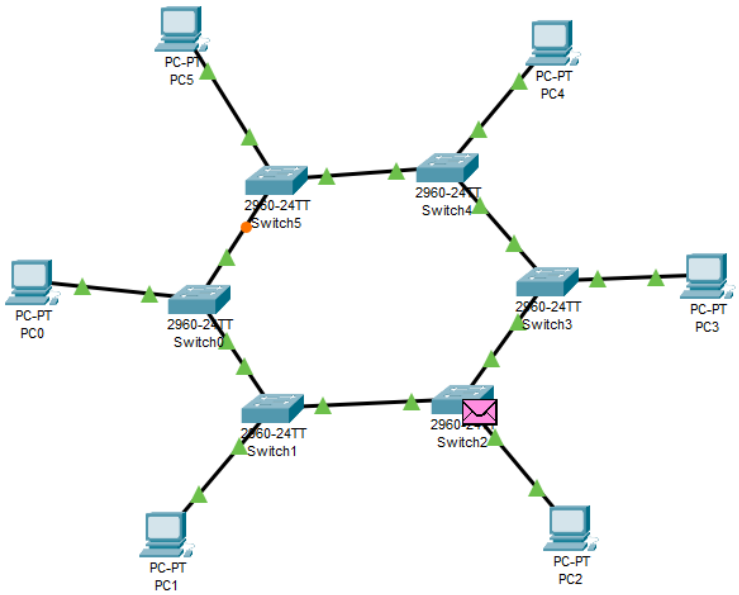


Figure 22: Simulation-mode view showing the ping frame being forwarded hop-by-hop through the switches around the ring toward its destination.

Observation:

The ping test succeeded with 0% packet loss. Unlike the hub topologies, Figure 22 shows the frame being passed hop-by-hop from switch to switch around the ring rather than being flooded everywhere at once, confirming MAC-address-table-based forwarding even in a ring layout. (Note: a physical ring of switches like this relies on Spanning Tree Protocol to prevent broadcast loops in a real deployment; that was outside the scope of this exercise.)

RESULT:

Topology	Device(s) Used	Network	Ping Result	Observed RTT
Star	1 Switches	192.168.1.0/24	Success, 0% loss	0 - 6 ms
Star	1 Hubs	192.168.1.0/24	Success, 0% loss	0 - 16 ms
Bus	1 Hubs	192.168.2.0/24	Success, 0% loss	0 ms
Ring	6 Switches	192.168.3.0/24	Success, 0% loss	0 ms

CONCLUSION:

In this Experiment, we successfully designed and configured four different LAN topologies, a star topology using a switch, a star topology using a hub, a bus-style topology using a hub, and a ring topology using switches in Cisco Packet Tracer. IP addresses were assigned to all connected PCs, and end-to-end connectivity was verified using the Ping command in every topology, with 0% packet loss recorded in all cases. The hub-based topologies visibly broadcast every frame to all connected stations, whereas the switch-based topologies forwarded frames only to the intended port, consistent with a switch's MAC-address-table-driven forwarding. From the experiment, we understood that switches provide more efficient, secure, and collision-free communication than hubs, making them the preferred choice for modern LANs regardless of the physical cabling topology used.